First Half

I ♡ 6103

	Financial calculation	Excel syntax
Annuity	Present value of annuity including a final payment of FV	=pv(rate, nper, pmt, [fv], [type])
	Payment that makes present value equal to target value	=pmt(rate, nper, pv, [fv], [type])
	Number of payments needed to match present value to target value	=nper(rate, pmt, pv, [fv], [type])
	Interest rate per period that matches present value and target value	=rate(nper, pmt, pv, [fv], [type])
Other	Final payment that matches present value and target value	=fv(rate, nper, pmt, [fv], [type])
	Present value of a range of cash flows starting in one period	=npv(rate, range of values)
	Rate of return that makes the present value of a stream of payments zero (the first payment is immediate)	=irr(range of values, [guess])

=normsinv() Inverse standard normal CDF function
=normsdist() Standard normal CDF function

$$PMT_0 = \frac{PMT}{1+g} \qquad P_{t+j} - \mu = (P_t - \mu)(1-\kappa)^j \qquad EAY = \left(\frac{100}{Price}\right)^{365/days} - 1$$

$$r^* = \frac{r-g}{1+g} \qquad (1+EAR) = (1+\frac{r}{k})^k \qquad PV = \frac{f_{t-1,t}F}{(1+r_{st})^t}$$

No Arbitrage: f is set to ensure (a) and (b) have the same FV

$$1.022^2 = 1.02(1+f) \rightarrow f = 2.4004\%$$

$$t = -\ln(1-u)/\ \text{lambda}$$

Discrete time discounting

$$\rho = \frac{r-i}{1+i}$$

Continuous discounting

$$\rho = r - i$$

$$\frac{F(t)}{X_{CNY}} = \frac{(1+r_{CNY})^t}{(1+r_{USD})^t} \equiv (1+\delta)^t$$

$$E[f(z)] = \bar{X} \pm 2s/\sqrt{N}$$

Bond (promised) yield = $r + k + \pi\lambda$
Credit spread = $k + \pi\lambda$

$$z_2 = \rho z_1 + z_3\sqrt{1-\rho^2}$$

$$V_0 = \frac{C}{r}\left(1 - \frac{1}{(1+r)^n}\right) \text{ for } r \neq 0$$

$$PVLGP(a,b,r,s) = \frac{\left(\frac{a+bs}{r} + \frac{b}{r^2}\right)}{(1+r)^{s-1}}$$

$$\text{Sharpe Ratio} = \frac{r_i - r}{\sigma_i} = \frac{Expected\ dollar\ return - r.f.funding\ costs}{Dollar\ standard\ deviation}$$

ABM

$$S_{t+1} = S_t + \mu + \sigma z_1$$
$$S_{t+j} = S_t + j\mu + \sigma\sum_{i=1}^{j} z_{t+i}$$
$$E[S_{t+j}|S_t] = S_t + j\mu$$
$$Var[S_{t+j}|S_t] = j\sigma^2$$
$$Cov[S_{t+j}, S_{t+m}|S_t] = \sigma^2 \min(j,m)$$

GBM

$$S_{t+j} = S_t e^{\left(\alpha - \frac{1}{2}\sigma^2\right)j + z_j\sigma\sqrt{j}}$$
$$E\left[ln(S_{t+j})\,|S_t\right] = ln(S_t) + \left(\alpha - \frac{\sigma^2}{2}\right)j$$
$$Var\left[ln(S_{t+j})\,|S_t\right] = \sigma^2 j$$
$$Cov\left[ln(S_{t+j}), ln(S_{t+m})|S_t\right] = \sigma^2\, min(j,m)$$

MR

$$S_{t+1} = S_t + \kappa(\mu - S_t) + \sigma z_1$$
$$S_{t+j} = \mu + (S_t - \mu)(1-\kappa)^j + \sigma\sum_{i=0}^{j-1}(1-\kappa)^i z_{t+j-i}$$
$$E[S_{t+j}|S_t] = \mu + (S_t - \mu)(1-\kappa)^j$$
$$Var[S_{t+j}|S_t] = \sigma^2\left(\frac{1-(1-\kappa)^{2j}}{\kappa(2-\kappa)}\right)$$
$$Cov[S_{t+j}, S_{t+m}|S_t] = \sigma^2\left(\frac{1-(1-\kappa)^{2j}}{\kappa(2-\kappa)}\right)(1-\kappa)^{m-j} \text{ for } m \geq j$$

$$A = 1'\Sigma^{-1}1$$
$$B = 1'\Sigma^{-1}r$$
$$C = r'\Sigma^{-1}r$$
$$D = AC - B^2$$
$$\sigma^2(R) = \frac{AR^2 - 2BR + C}{D}$$

$$r_T = \frac{C - Br_f}{B - Ar_f}$$
$$w_T = \frac{\Sigma^{-1}(r - r_f 1)}{B - Ar_f}$$

$$s^2(X_h'b) = s^2 X_h'(X'X)^{-1}X_h$$

Second Half

Table 1

Derivation of the Equilibrium Valuation of n Correlated Cash Flows in the 1-period CFPT Model

	Variance Utility	Standard Deviation Utility
Expected wealth	N′μ + (W₀ – N′V)(1+r)	
Variance of wealth	N′ΣN	
Expected utility	N′μ + (W₀ – N′V)(1+r) - ½AN′ΣN	N′μ + (W₀ – N′V)(1+r) – A(N′ΣN)^½
First order condition	μ – V(1+r) – AΣN = 0	μ – V(1+r) – AΣN/(N′ΣN)^½ = 0
Valuation with market clearing (N=1)	$V = \frac{\mu - A\Sigma 1}{(1+r)}$	$V = \frac{\mu - A\Sigma 1/(1'\Sigma 1)^{1/2}}{(1+r)}$
GVE	$rV + A\Sigma 1 = \mu - V$	$rV + \frac{A\Sigma 1}{(1'\Sigma 1)^{1/2}} = \mu - V$
Aggregate A	$A = \frac{1'(\mu - V(1+r))}{1'\Sigma 1}$ "Market price of variance risk"	$A = \frac{1'(\mu - V(1+r))}{(1'\Sigma 1)^{1/2}}$ "Dollarized Market Sharpe Ratio"

$$V_C = \left(V - \frac{X}{1+r}\right)N(d^*) + \frac{\sigma n(d^*)}{1+r} = \frac{\mu_C^*}{1+r}$$

$$d^* = \frac{(1+r)V - X}{\sigma}$$

$$C_0 = S_0 N(d_1) - X e^{-rT} N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S_0}{X}\right) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}} \qquad d_2 = d_1 - \sigma\sqrt{T}$$

$$F_{0,T} = S_0 \exp[(r-\delta)T]$$

Table 2

Risk Premium Calculations in the Single Period Variance Utility Model

Calculation	Dollars	Returns (percentages)
Total expected return	$\mu - V$	$\frac{\mu - V}{V} = (r\mu + A\Sigma 1) \oslash (\mu - A\Sigma 1)$
Risk premium	$\mu - (1+r)V = A\Sigma 1$	$[\mu - (1+r)V] \oslash V = A\Sigma 1 \oslash V$
Covariance with market	$\Sigma 1$	$\frac{\Sigma 1 \oslash V}{1'V}$
Variance of market	$1'\Sigma 1$	$\frac{1'\Sigma 1}{(1'V)^2}$
Beta	$\frac{\Sigma 1}{1'\Sigma 1}$	$\frac{\Sigma 1 \oslash V}{1'\Sigma 1} 1'V$
Risk premium⊘Beta (constant for all securities)	$A1'\Sigma 1$	$\frac{A1'\Sigma 1}{1'V}$

•The symbol "⊘" is being used to represent the Hadamard quotient or the Schur quotient, which is the element-by-element quotient of two vectors or matrices

$$r_m = \frac{1'(\mu - V)}{1'V} \qquad r_i = \frac{\mu_i - V_i}{V_i} \qquad \beta_i = \frac{\Sigma_i 1}{1'\Sigma 1} \cdot \frac{1'V}{V_i}$$

$$r_i - r = \beta_i (r_m - r)$$

- Partition the cash flows into known [1..j] and unknown [j+1..N] cash flows at any point in time j
- Then we have $C \equiv \begin{pmatrix} C_j \\ C_N \end{pmatrix}$ $\mu \equiv \begin{pmatrix} \mu_j \\ \mu_N \end{pmatrix}$ $\Sigma \equiv \begin{pmatrix} \Sigma_{jj} & \Sigma_{jN} \\ \Sigma_{Nj} & \Sigma_{NN} \end{pmatrix}$
- This gives us a kind of multiple regression formula together with the Schur decomposition of the covariance matrix
  - $E(C_N|C_j) = \mu_N + \Sigma_{Nj}\Sigma_{jj}^{-1}(C_j - \mu_j)$
  - $\Sigma_j = \Sigma_{NN} - \Sigma_{Nj}\Sigma_{jj}^{-1}\Sigma_{jN}\ for\ j = 1..n-1\ with\ \Sigma_0 = \Sigma$
  - (with the single subscript, the Σ is the covariance matrix for the unknown cash flows after time j, given the cash flows up to time j)

Variance formula

$$V_j = B'E[C_N|C_j] - kB'(1'\Sigma_j 1 - 1'\Sigma_{j+1}1)$$

Standard deviation formula

$$V_j = B'E[C_N|C_j] - kB'(sqrt(1'\Sigma_j 1) - sqrt(1'\Sigma_{j+1}1))$$